Ch 6.1: Subset Selection

Lecture 16 - CMSE 381

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Announcements

Last time

- k-fold CV for Classification

Covered in this lecture

- response to feedback (from Quiz 5)
- Subset selection
- Forward and Backward Selection

Announcements:

- HW #4 Due Sunday (10/12)
- HW #5 due very late, to avoid Fall break weekend. I will release it early so you can work on it to prep for the exam.

CMSE381_F2025_Schedule : Schedule

	M	9/22	***Project Day & Review***		
	W	9/24	**Midterm #1**		
12	F	9/26	Leave one out CV	5.1.1, 5.1.2	
13	M	9/29	k-fold CV	5.1.3	
14	W	10/1	More k-fold CV	5.1.4-5	
15	F	10/3	k-fold CV for classification	5.1.5	
16	M	10/6	Subset selection	6.1	
17	W	10/8	Shrinkage: Ridge	6.2.1	
18	F	10/10	Shrinkage: Lasso	6.2.2	HW #4 Due Sun 10/12
19	M	10/13	PCA	6.3	
20	W	10/15	PCR	6.3	
	F	10/17	***Review***		
	M	10/20	Fall Break		
	W	10/22	**Midterm #2**		
21	F	10/24	Polynomial & Step Functions	7.1-7.2	HW #5 Due Sun 10/28
22	M	10/27	Step Functions; Basis functions; Start Splines	7.2-7.4	
23	W	10/29	Regression Splines	7.4	

Feedback collected in Quiz 5

- what I can do:
  - I will add overview slide at the beginning of each lecture. This will become a study guide for the exam.
  - I will provide you a practice exam with matched difficulty. Those questions will not be in the exam.
  - Code portfolios as bonus assignments. I will provide a template.
- what you can do:
  - Take better notes
  - Read the textbook (read before class if you have trouble following the lecture)
  - Ask more questions (in class, on slack, office hours/helproom) and respond to the review class "burning questions" survey

What should you learn from this lecture?

- Why shouldn't you have as many predictors in your model as possible?
- What are the three basic methods could be used to select the appropriate predictors (feature selection)?
- What are the steps/procedures in the algorithm for each of these methods?
- How to implement these procedures by hand given the training and CV test error for each combo of predictors?
- What are the pros and cons for choosing each of these feature selection methods? When you can or cannot use them?
- How do you implement the feature selection algorithm in Python?

Section 1

Previously on linear regression ...

The problem of many features (p) relative to samples (n)

Up to now, we've focused on standard linear model: $Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$ and done least squares estimation. *Translation: Minimize RSS* $= \sum_i (y_i - \hat{y}_i)^2$

Prediction accuracy

- If the true relationship is approximately linear, least squares estimates have low bias
- If $n \gg p$, then least squares also has low variance
- If n not much larger than p, then high variability, overfitting, poor predictions
- If $n < p$ then no unique solution so can't use at all
- Goal on next slide: shrink/constrain number of variables to improve accuracy

The problem of many features (p) relative to samples (n)

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Model Interpretability

- Often, many variables included aren't associated with response
- Including them leads to unnecessary complexity in the model
- Idea: Set these coefficients to 0 (or close to zero)
- Goal on next slide: Do some automatic feature selection / variable selection to get rid of unnecessary variables

Section 2

Best Subset Selection

Go through each combo of variables exhaustively (exhausting?)

All subsets of 4 variables ($2^4 = 16$)

- $\emptyset$
- X_1
- X_2
- X_3
- X_4
- $X_1\ X_2$
- $X_1\ X_3$
- $X_1\ X_4$
- $X_2\ X_3$
- $X_2\ X_4$
- $X_3\ X_4$
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- The game: fit the model 2^n times, score each, pick one with best score
- point out this is tiered by number of variables (k) out of p variables
- easy to compare model with the same k (no order difference)
- after come back from next slide: each tier has p choose k combos $\binom{p}{k} = \frac{p!}{k!(p-k)!}$

One way of breaking this up

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
   (b) Pick the best among these $\binom{p}{k}$ models, and call it $\mathcal{M}_k$. Here *best* is defined as having the smallest RSS, or equivalently largest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

- Part 2b goes for lowest training score, Part 3 then goes for lowest testing score.
- Go back to previous page and walk through the algorithm and variables slowly
- point out that Step 2 is computational infeasable for large p

Calculate by hand

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the MSE value computed for the model learned using each possible subset of variables.

	Training MSE (x10^7)	k-fold CV Testing Error
Null model	8.76	10.08
X1	8.63	9.98
X2	7.42	8.01
X3	8.16	8.3
X4	8.33	9.06
X1,X2	4.33	7.47
X1,X3	5.82	5.22
X1,X4	3.17	4.23
X2,X3	4.07	3.78
X2,X4	3.31	4.01
X3,X4	3.06	4.16
X1,X2,X3	3.08	5.49
X1,X2,X4	3.55	4.02
X1,X3,X4	2.97	4.23
X2,X3,X4	2.98	3.17
X1,X2,X3,X4	2.16	4.39

1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using best subset selection?
2. What subset of variables is returned using best subset selection?

Extra work space if it helps

	Training MSE (x10^7)	k-fold CV Testing Error
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$\mathcal{M}_0 = \emptyset, \mathcal{M}_1 = \{X_2\}, \mathcal{M}_2 = \{X_3, X_4\}, \mathcal{M}_3 = \{X_1, X_3, X_4\}, \mathcal{M}_4 = \{X_1, X_2, X_3, X_4\}, .$

Among test error, best is X_3X_4

Code to do this

Section 3

Forward Selection

What's the problem with best subset selection?

- Checking 2^p models is not reasonable for large p, $p > 40$
- The next bits are finding alternatives to Step 2

Forward Stepwise Selection

Algorithm 6.2 *Forward stepwise selection*

1. Let $\mathcal{M}_0$ denote the *null* model, which contains no predictors.
2. For $k = 0, \ldots, p-1$:
   (a) Consider all $p-k$ models that augment the predictors in $\mathcal{M}_k$ with one additional predictor.
   (b) Choose the *best* among these $p-k$ models, and call it $\mathcal{M}_{k+1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

An example for Forward Stepwise Selection

- $\emptyset$

- X_1
- X_2
- X_3
- X_4

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Group work: by hand same example with forward example

We train a model using four variables, X_1, X_2, X_3, X_4. We're interested in getting a subset of the variables to use. The following table shows the mean squared error and the R^2 value computed for the model learned using each possible subset of variables.

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1. What subset of variables is found for each of the sets $\mathcal{M}_0, \mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3, \mathcal{M}_4$ when using forward selection?
2. What subset of variables is returned using forward subset selection?

Extra work space if it helps

	Training MSE (x10^7)	k-fold CV Testing Error
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$\mathcal{M}_0 = \emptyset, \mathcal{M}_1 = \{X_2\}$, $\mathcal{M}_2 = \{X_2, X_4\}$, $\mathcal{M}_3 = \{X_2, X_3, X_4\}$, $\mathcal{M}_4 = \{X_1, X_2, X_3, X_4\}$, .

Among test error, best is $\{X_2, X_3, X_4\}$

Pros and Cons of Forward Stepwise

Pros:

- Computationally cheaper
- Number of models fit is

$$1 + \sum_{k=0}^{p-1}(p-k) = 1 + \frac{p(p+1)}{2}$$

which is way better than 2^p

Cons:

- Not guaranteed to find the best model
- As example: if best 1-variable model is X_1, but best 2-variable model is X_2X_3, then forward selection won't find it.
- Is this a con? Maybe just a limitation If $n < p$, then can only construct models $\mathcal{M}_0 \cdots \mathcal{M}_{n-1}$

Section 4

Backward Selection

Backward stepwise selection

Algorithm 6.3 *Backward stepwise selection*

1. Let $\mathcal{M}_p$ denote the *full* model, which contains all p predictors.
2. For $k = p, p-1, \ldots, 1$:
   (a) Consider all k models that contain all but one of the predictors in $\mathcal{M}_k$, for a total of $k-1$ predictors.
   (b) Choose the *best* among these k models, and call it $\mathcal{M}_{k-1}$. Here *best* is defined as having smallest RSS or highest R^2.
3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

Pros and Cons of Backward Stepwise

Pros:

- Computationally cheaper
- Number of models fit is still

$$1 + \sum_{k=0}^{p-1}(p-k) = 1 + \frac{p(p+1)}{2}$$

which is way better than 2^p

Cons:

- Not guaranteed to find the best model
- Unlike forward selection, this can't be used at all if $n < p$

Can also look at hybrid approaches with both forward and backward selection

TL;DR

Algorithm 6.1 *Best subset selection*

1. Let $\mathcal{M}_0$ denote the *null model*, which contains no predictors. This model simply predicts the sample mean for each observation.
2. For $k = 1, 2, \ldots p$:
   (a) Fit all $\binom{p}{k}$ models that contain exactly k predictors.
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3. Select a single best model from among $\mathcal{M}_0, \ldots, \mathcal{M}_p$ using cross-validated prediction error, C_p (AIC), BIC, or adjusted R^2.

- Modify step 2 with forward or backward selection
- Choose best model in step 3 using one of our adjusted training scores or CV

Test your understanding: PollEv

Next time

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